

# Labour Supply 3: Labour Supply Over the Life Cycle

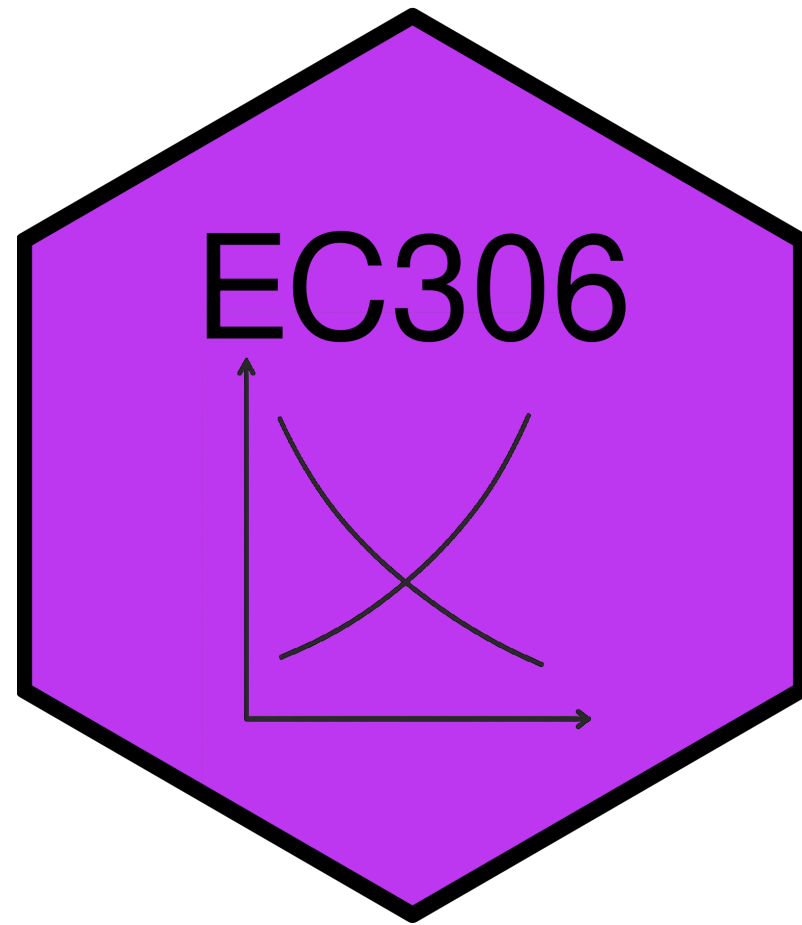
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EC306- Wages and Employment

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# Goals of This Section

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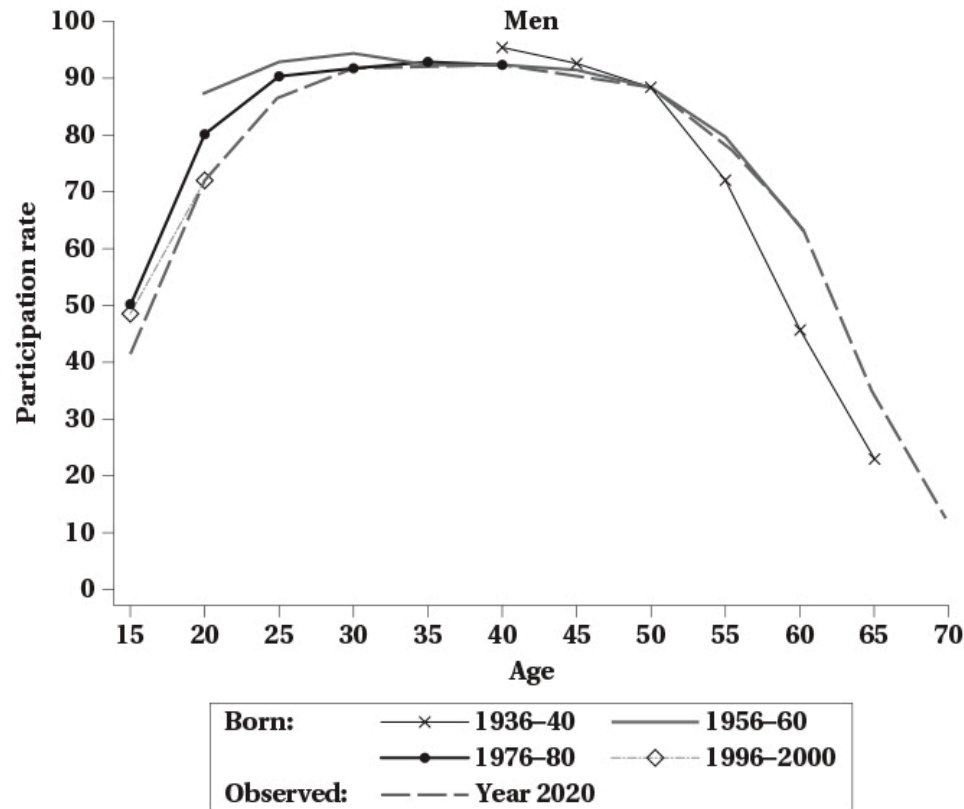
- Show patterns of labour supply over a person's lifetime
- Discuss issues comparing labour supply of people at different ages
- Introduce lifetime labour supply model
- Discuss household production
- Analyze retirement decisions

# Life Cycle Labour Supply

# Introduction

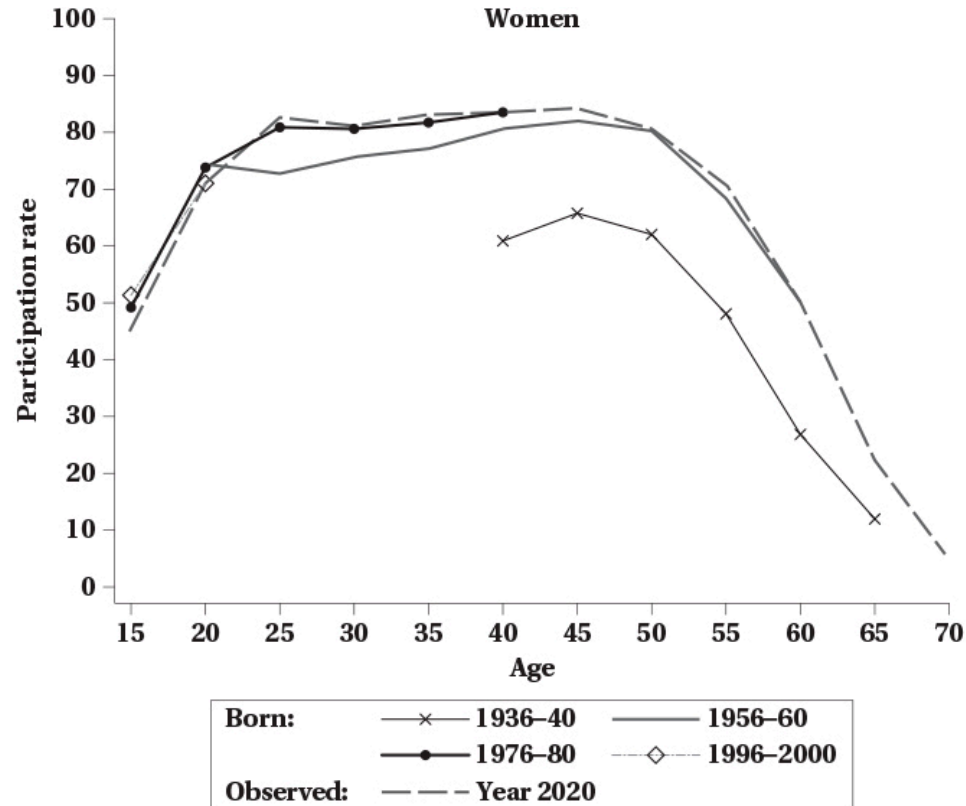
- Previous discussion focused on labour supply at a moment in time
  - Labour supply changed immediately in response to changes in wages, non-labour income, or policies
- In reality, people can plan work over their lifetime
  - Participation is highest in prime working years
  - Life events alter labour supply decisions now and in the future
    - Marriage, children, education, health, retirement
- A life cycle approach helps us understand these patterns

# Life Cycle Participation for Men



- Graph follows labour supply of men born in different years
- Participation rises quickly in early 20s, peaks in late 40s, then declines
- There are differences for different birth cohorts
  - At younger ages, younger cohorts participate a bit less
  - At older ages, younger cohorts participate more

# Life Cycle Participation for Women



- Women follow similar lifetime pattern
- Rates are shifted down a bit relative to men
- Participation higher for younger cohorts at all ages
  - Consistent with rising female labour force participation over time

# Cohort Effects vs Age Effects

- Previous graphs highlight a technical point
- Each graph contains a line that looks at life cycle pattern in year 2020
  - Participation of people in 2020 at different ages
- Problem: people at different ages are born at different times
  - 20 year olds in 2020 born in 2000
  - 60 year olds in 2020 born in 1960
- So when we look at participation by age in a single year, we are mixing age effects and cohort effects
  - Age effects: how labour supply changes as people get older
  - Cohort effects: how labour supply differs for people born at different times



# Cohort Effects vs Age Effects

- Cohort effects can be significant
  - In particular for women where it has been increasing for younger cohorts
- It is not a good approach to look at participation by age in a single year and interpret as pure age effects
- If pure age effects are of interest, better to look at participation by age for a single birth cohort
  - This is done in the graphs on the previous two slides
- Notice that the lines shift up and down for different birth cohorts
- Pattern across age is similar but there are some differences

# Cohort Effects vs Age Effects

- Practically, it is difficult to do this
- Very few datasets have data on the same people over their entire life
  - Hard to track people over time
  - People die, move, refuse to respond
  - Data that does track people are often restricted access
- Another approach is to use **quasi-cohorts**
  - Group people by birth year in cross-sectional (single year) data
  - Look at how participation changes for these groups over time
  - This is what the graphs on the previous two slides do

# Cohort Effects vs Age Effects

- You can do this with the Canada Census
  - Census is collected every 5 years
- Each Census collects information on age and labour force status (among many other things)
- Can follow a birth cohort across different Census years as they get older

| Birth Cohort | Census Year |      |      |      |      |
|--------------|-------------|------|------|------|------|
|              | 1981        | 1986 | 1991 | 1996 | 2001 |
| 1950         | 31          | 36   | 41   | 46   | 51   |
| 1951         | 30          | 35   | 40   | 45   | 50   |
| 1952         | 29          | 34   | 39   | 44   | 49   |
| 1953         | 28          | 33   | 38   | 43   | 48   |
| 1954         | 27          | 32   | 37   | 42   | 47   |
| 1955         | 26          | 31   | 36   | 41   | 46   |

# Cohort Effects vs Age Effects

- One caveat is that in each Census year, people in a birth cohort are different people
  - Not following the exact same individuals over time
- This is generally accepted as a reasonable approximation

# Cohort Effects vs Age Effects

## Born in 1975

| Observed in        | 2000 | 2005 | 2010 |
|--------------------|------|------|------|
| Age                | 25   | 30   | 35   |
| Participation Rate | 75%  | 80%  | 85%  |

## Born in 1980

| Observed in        | 2000 | 2005 | 2010 |
|--------------------|------|------|------|
| Age                | 20   | 25   | 30   |
| Participation Rate | 75%  | 80%  | 85%  |

- To illustrate problem with following ages in single Census year, examine this table
- Tracks birth cohorts across Census years
- Shows age and participation rate
- For each cohort participation rises 5 points per year
- But looking at ages in a single Census year, they appear not to change

# Cohort Effects vs Age Effects

- Where do cohort effects come from? Why do we expect cohorts to be different?
- People in part are defined by their generation
  - Social attitudes change over time
  - Economic conditions and policies change over time
  - Generations experience different significant events (e.g. wars, disease)
- We therefore expect cohorts to differ in their attitudes and behaviours
  - Research has shown women born in early 1930s had higher labour supply than previous generations
  - They were influenced by their mothers working during war
  - Male children also more likely to be married to working women later in life
- Important to separate these from effects of pure aging

# Dynamic Life Cycle Models

# Introduction

- We now know that labour supply varies over the life cycle
- The allocation of work over time can be modeled
- Models examine the effects on labour supply of wage changes
  - Wages can change at different ages
  - They can change for a short time or permanently
  - They can be anticipated or unanticipated
- Different types of changes have different effects on labour supply



# Model

- Individuals will still maximize utility over consumption and leisure subject to a budget constraint
- The utility is a function of consumption and leisure over the lifetime

$$u = U(C_1, C_2, \dots, C_N, L_1, L_2, \dots, L_N)$$

- Here  $U(.)$  is a function
  - e.g. could be logarithmic, cobb-douglas, etc.
- To define the budget constraint, recognize that we can consume and work in each period
- In theory, we can consume more than our income in a period by borrowing
  - We can also save some income to consume later
- We therefore need to be conscious of the interest rate on saving and borrowing

# Model

- Consider a two period model
- The budget constraint says that the present value of consumption equals the present value of income

$$C_1 + \frac{C_2}{1+r} = W_1H_1 + \frac{W_2H_2}{1+r}$$

- $r$  is the interest rate for borrowing and saving
- This is the present (period 1) value formulation of the lifetime budget constraint
- You can also express it in future (period 2) value

$$(1+r)C_1 + C_2 = (1+r)W_1H_1 + W_2H_2$$

# Model

- The ability to borrow and save means a person can consume more than their income in a period
- For example, rearranging the future value of the budget we get

$$C_2 = (1 + r)(W_1H_1 - C_1) + W_2H_2$$

- If  $W_1H_1 > C_1$ , then the person saved some income in period 1 to consume in period 2
  - That savings grows with interest, and they have  $(1 + r)(W_1H_1 - C_1)$  to spend in period 2
- If  $W_1H_1 < C_1$ , then the person borrowed to consume more than their income in period 1
  - $(1 + r)(W_1H_1 - C_1)$  is negative and it reduces their income in period 2

# Model

- In our model, the choice is between consumption and leisure
- To reexpress this in terms of leisure, note that  $H_t = T - L_t$ , so

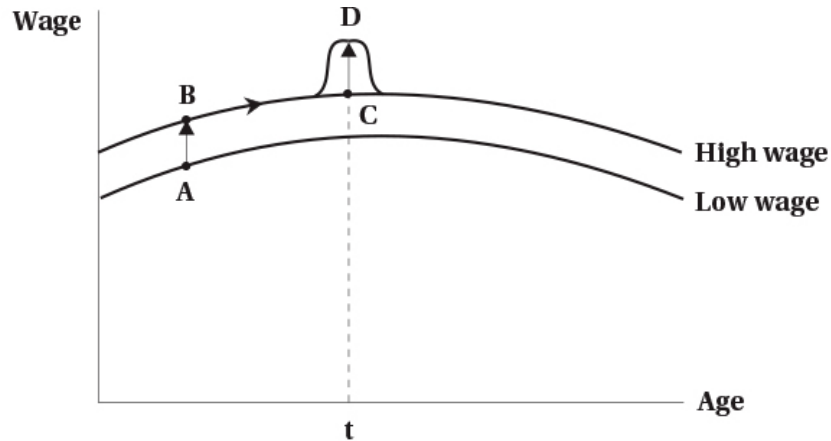
$$C_1 + \frac{C_2}{1+r} + W_1 L_1 + \frac{W_2 L_2}{1+r} = W_1 T + \frac{W_2 T}{1+r}$$

- You can consume two goods (consumption and leisure)
- The price of consumption is 1 in both periods
- The price of leisure is  $W_1$  in period 1 and  $W_2$  in period 2
- You consume these out of “full income”
  - Full income is the income you would have if you worked all available hours in both periods
- Period 2 values are discounted by the interest rate

# Model

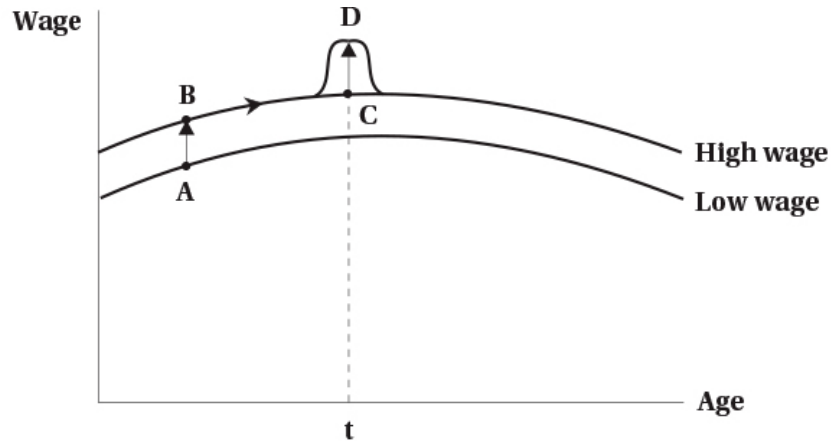
- If you maximize utility subject to the budget constraints, you get a labour supply function
  - Tells you optimal labour supply as a function of wages, interest rates, lifetime resources
- Like previous models, we can examine effects of wage changes
- There are three types
  - A permanent unanticipated change in the wage in all periods
  - An anticipated future change in the wage
  - A temporary unanticipated change in the wage in one period only
- These lead to substitution and income effects
  - More complicated because of the extra time dimension of the model
- We can see these effects in graph form

# Permanent Wage Change



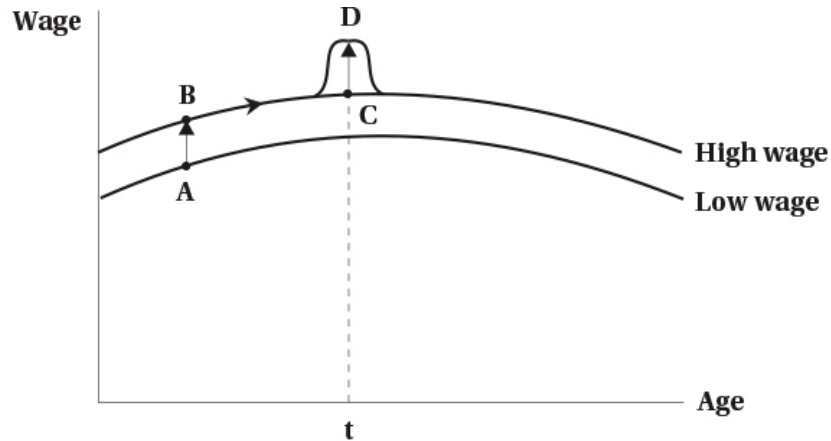
- Consider first a permanent, unanticipated change in the wage
  - Increases wage in all periods
  - Shifts wage schedule up
- Higher wage has two effects
  - Income effect: higher wage means higher lifetime income, so person consumes more leisure (less work)
  - Substitution effect: higher wage means leisure is more expensive, so person consumes less leisure (more work)
- Labour supply effect is ambiguous

# Anticipated Wage Change



- An anticipated wage change moves person along their wage schedule
- There is no income effect in this case
  - Person optimized over their lifetime income
  - Lifetime income has not changed
- The substitution effect shifts leisure towards periods where wage is lower
  - If wage is higher in period 2, leisure is more expensive in period 2
  - Less leisure in period 2 and more in period 1
  - Person works more in period 2 and less in period 1

# Temporary Unanticipated Wage Change



- A temporary unanticipated wage change alters wage for one period (or a small number of them) only
  - e.g. bonus pay, overtime pay
- Creates a temporary blip upward in wage schedule
- Two effects
  - Small Income effect: slightly higher lifetime income, so person consumes more leisure (less work)
  - Substitution effect: leisure is more expensive in the period with the higher wage, so person consumes less leisure (more work)



# Comparing Effects of Wage Changes

- Strongest effect on labour supply is the anticipated wage change
  - Pure substitution effect
  - No offsetting income effect
- Next strongest is the temporary unanticipated wage change
  - Substitution effect
  - Small income effect
  - Person works more in period with higher wage
- Permanent unanticipated wage change has ambiguous effect
  - Substitution effect
  - Larger income effect
  - Ambiguous effect on labour supply

# Evidence on Life Cycle Labour Supply

- Interesting experiment on bike messengers in Zurich
- Messengers paid a percentage of daily revenues
- Can choose their own total work hours
- Researchers randomly assigned messengers to
  - 25% higher wage for one month in September of one year (group A)
  - Others to 25% higher wage for one month in November of the same year (group B)
- Each acted as control group for the other
- Both get same increase in lifetime income, so difference is substitution effect
- Showed messengers shift labour supply to periods with higher wage

# Household Production

# Introduction

- So far we assumed that people choose between consumption and leisure
  - Leisure is time not working for pay
  - Encompasses pure leisure, household production, and other non-market activities
- In reality, people also spend time on household production
  - Cooking, cleaning, childcare, home maintenance, shopping, etc.
- Household production produces goods and services that can be consumed
  - Meals, clean clothes, child supervision, etc.
- In this section we explicitly model household production

# Model

- Assume that people gain utility from the consumption of goods and services

$$u = f(Z_1, Z_2, \dots, Z_N)$$

- In this model
  - $Z_1, Z_2, \dots, Z_n$  are final goods and services
  - $f(.)$  is a function (e.g. cobb-douglas, logarithmic, etc.)
- Each of these goods and services are produced by the household using
  - A set of market inputs  $X_1, X_2, \dots, X_n$
  - Time  $h_i$
- In this setup, time is divided between
  - Market work
  - Household production tasks



# Model

- What does it mean that households “produce” the goods they consume?
- Does not mean that literally everyone is producing goods from raw materials
- Instead means that every good/service consumed requires some time to prepare
  - Hiring contractors to remodel a home requires a small amount of home production
    - Time spent explaining what you need, keeping contractor on schedule
  - Remodelling your basement by yourself requires much more time
- In this model we explicitly account for substituting time between various uses
  - Work, home production of various goods, leisure

# Model

- Example: a household consumes two goods, nutrition and rest
  - These would be  $Z_1$  and  $Z_2$  in the utility function
- Nutrition is produced using processed or raw food, and time
  - These are  $X_1$ ,  $X_2$  and  $h$
  - Raw food takes time to prepare equal to  $h = d$  hours
- Each of the three purchased goods has a price
- Processed food
  - Price per unit of nutrition is  $P_c$
  - Takes no time to prepare

# Model

- Raw food
  - Price per unit of nutrition is  $P_r + W_d$
  - The added time cost is valued at the wage rate  $W$  times  $d$  hours
- Leisure
  - Price is the wage rate  $W$



# Comparative Statics

- What happens when we start changing some of the prices?
- Increase in wage rate  $W$ 
  - Cost of time affects both rest and production of raw food
  - Higher wage makes home production more expensive
    - Time spent cooking becomes more expensive
    - Substitute away from home production to processed food
    - Substitution effect: less home production, more processed food
    - Income effect: higher wage means higher income, so more consumption of all goods
  - Effect on rest is same as before
    - Substitution effect: less rest, more work
    - Income effect: higher income, so more rest

# Comparative Statics

- Decrease in price of processed food  $P_c$ 
  - Processed food becomes cheaper relative to other goods
  - Substitution effect: more processed food, less home production
  - Income effect: lower price of processed food means higher real income, so more consumption of all goods
- Decrease in price of raw food  $P_r$ 
  - Raw food becomes relatively cheaper
  - Substitution effect: more raw food, less processed food
  - Income effect: lower price of raw food means higher real income, so more consumption of all goods

# Comparative Statics

- Decrease in time required to prepare raw food  $d$ 
  - Example: technology that reduces cooking time (e.g. Instant Pot)
  - Home production becomes relatively cheaper (through lower time cost)
  - Substitution effect: more raw food, less processed food
  - Additional substitution effect: higher real wage (i.e. they keep more of their wage), less rest, more work
  - Income effect: lower time cost of raw food means higher real income, so more consumption of all goods

# Discussion

- Model adds more real world complexity to match what people actually do
- Emphasizes that people do not just choose between work and leisure
- Also emphasizes the role of the household relative to the individual
  - Household members can specialize in different tasks
  - Changes for one household member can affect others
- Discussion above did not mention household production function, but it is important
  - Different households have different abilities to produce goods
  - This can affect their choices and responses to changes in prices

# Fertility and Childbearing

# Introduction

- Fertility is a key life event that affects labour supply
  - Particularly for women, but also men to some degree
- Children are typically planned to some degree
  - Timing of children is often chosen to fit with work plans
  - Number of children is often chosen based on lifetime work plans
- Economists have developed models to analyze fertility decisions
  - These models are similar to the life cycle labour supply model
- We will not go over the model because it is a bit beyond the scope of this course
- But we will discuss some of the key variables that affect fertility decisions

# Variables Affecting Fertility - Income

- Income is positively related to the number of children
  - Higher income means higher ability to pay for children
  - Children are a “normal good”
- Some disagreement on whether children are considered a normal good in our models
  - Empirically, higher income families and countries have lower fertility rates
  - Explanation is that substitution effect of higher income dominates income effect
    - Higher income means higher opportunity cost of time spent on children

# Variables Affecting Fertility - Price of Children

- Like goods, the demand curve for children slopes down
  - Higher “price” of children means fewer children are chosen
- Several elements involves in the price of children
  - Direct costs: food, clothing, housework, etc.
  - Opportunity costs: time spent caring for children could be spent working for pay
- A higher price (wage rate) creates income and substitution effects
  - Income effect: higher price means higher lifetime income, so more children
  - Substitution effect: higher price means time with children is more expensive, so fewer children
- Effect is theoretically ambiguous, but empirically substitution effect dominates



# Variables Affecting Fertility - Price of Related Goods

- Many goods are consumed as part of having children
  - Childcare, education, etc.
- These are complementary goods
  - If you have more children, you will consume more of these goods
  - You could also consider these direct costs of children
- We know that an increase in the price of a complementary good reduces demand
  - Higher price of childcare or education means fewer children

# Variables Affecting Fertility - Tastes

- Tastes for children vary across people and over time
  - Some people really want children, others do not
  - Some cultures value children more than others
  - Attitudes towards children have changed over time
- A stronger taste for children means more children
  - This is like a shift in the demand curve for children
- Over time in Canada, there has been a shift away from children
  - Fertility rates have declined over time
  - People are having fewer children than in the past

# Variables Affecting Fertility - Technology

- Technology in this case could mean several things
  - Could refer to the “production” of children
  - Might also relate to the goods and services used with children
    - Home automation, disposable diapers, baby monitors, carriers
- Some key developments in the last few decades
  - Birth control: allows people to better plan timing and number of children
  - Fertility treatments: allows some people to have children who otherwise could not
  - Medical care: reduces infant mortality, so fewer children needed to ensure some survive
  - Home technology: reduces time cost of childcare, so lower opportunity cost of children
- Higher technology in any of these areas tends to increase demand for children

# Empirical Evidence

- Economists have examined the relationship between income and fertility
- Difficult because of omitted variables
  - A simple correlation between income and fertility tends to be negative
  - Important to hold constant other variables that affect both income and fertility
    - Example: education levels are positively related to income and negatively related to fertility
- When other variables are held constant, income tends to be positively related to fertility
  - Consistent with children being a normal good
- Tax incentives also tend to increase fertility

# Retirement and Pensions

# Introduction

- Retirement is a key life event that affects labour supply
  - Most people retire at some point
  - Retirement age has been changing over time
- It can mean many things
  - Permanent exit from the labour force
  - A reduction to part-time work
  - Retiring from one full-time job and taking another
- For most people retirement means permanent exit from the labour force

# Determinants of Retirement - Age

- Age is the most important determinant of retirement
  - Most people retire between ages 60 and 70
  - Very few people retire before age 50 or after age 75
- Some places have mandatory retirement
  - In Canada it is generally illegal to force retirement based on age
  - Except if deemed a bona fide occupational requirement (BFOR), like for pilots, police, firefighters, air traffic controllers
  - Still exists in other parts of the world (e.g. Japan, China, parts of Europe)

# Determinants of Retirement - Wealth/Earnings

- Wealth in our model entered as non-labour earnings
  - Higher wealth means higher non-labour income
  - Higher non-labour income means more leisure (less work)
- Earnings changes have income and substitution effects
  - Income effect: higher earnings means higher lifetime income, so more leisure (less work)
  - Substitution effect: higher earnings means leisure is more expensive, so less leisure (more work)
- Can also think of this in a lifecycle. model



# Determinants of Retirement - Other

- Health is an important determinant of retirement
  - Poor health can force people to retire early
  - Good health can allow people to work longer
  - Health of family members can also play a role
  - Might operate through non-labour income or preferences
- Changing nature of work can influence decisions
  - Move towards less physically demanding jobs can prolong work life
  - More flexible work arrangements can allow people to work longer
  - Alternatively, more stressful jobs can lead to earlier retirement
  - Automation might force early retirement
  - Could affect the budget constraint or preferences

# Determinants of Retirement - Other

- Changing nature of family
  - There are many more two-earner families now
  - This can lead to earlier retirement for one spouse
  - Or could prolong work if spouses want to work together longer
  - Children living at home for longer can also affect retirement timing
  - Increase in caring for elderly parents can lead to earlier or later retirement
  - Might affect budget constraint or preferences

# Retirement Income

- Public and private pension plans have major effect on retirement decisions
  - They determine most income in retirement
- The pension system in Canada consists of

## 1. Old Age Security (OAS)

- A flat rate pension available to most Canadians over age 65
- Funded out of general tax revenues
- Monthly max is \$740 for those aged 65-74, \$814 for those 75+
- Clawback for high income earners (above \$148,000 in 2024)
- Supplemented by the Guaranteed Income Supplement (GIS) for low income seniors

# Retirement Income

## 2. Canada/Quebec Pension Plan (CPP/QPP)

- Earnings-related pension available to those who have contributed
- Funded by contributions from employees, employers, and self-employed
- Monthly max is \$1,433 in 2025 (age 65)
- Can start receiving as early as age 60 (reduced) or as late as age 70 (increased)
- Based on average earnings over working life, up to a maximum limit

## 3. Employer Pension Plans

- Many employers offer pension plans to employees
- Can be defined benefit or defined contribution plans
- Vary widely in terms of benefits and eligibility
- Includes private savings from RRSPs

# Old Age Security

- What effect does OAS have on retirement decisions?
- Operates as a lump sum transfer
  - Increases non-labour income
  - Creates income effect: more leisure (less work)
  - No substitution effect because it does not change the wage rate
- GIS operates like a negative income tax
  - People get a the lump sum payment
  - Payments reduced by \$0.50 for each dollar of work income
  - Has work incentives equal to a negative income tax

# Canada/Quebec Pension Plan

- The effect of pension income on retirement depends on how it is structured
- In Canada currently, it is simply a fixed payment
  - Increases non-labour income
  - Creates income effect: more leisure (less work)
  - No substitution effect because it does not change the wage rate
- In the past, there was a provision that reduced benefits if you worked
  - Similar to a negative income tax
  - If the tax back rate is 100%, then it is like pure welfare
  - Could also not tax back until a certain income level

# Employer Pension Plan

- Generally involve a fixed payment based on years of service and earnings
  - Increases non-labour income
  - Creates income effect: more leisure (less work)
  - No substitution effect because it does not change the wage rate
- Over time the parameters that determine the fixed payment can change
  - This can alter work incentives at retirement
  - Example: if the benefit formula is changed to require more years of service
    - Would reduce the income effect of the pension
    - Could also create a substitution effect to work more to get the pension

# Evidence on Effect of Pensions

- Studies find that pensions affect retirement in expected ways
  - Pensions that increase non-labour income lead to earlier retirement
  - Pensions that create work incentives lead to later retirement
- More recent work
  - Increase in retirement age increases working-age disability claims in UK
  - Increasing public pension participation improves health behaviours (less smoking)
  - Pension reform allowing Japanese spouses to claim half of pension earnings increased divorce



# Summary

# Summary

- Labour supply varies over the life cycle
  - Participation highest in prime working years
- A life cycle approach helps us understand these patterns
- Dynamic life cycle models can analyze effects of wage changes
  - Permanent unanticipated changes
  - Anticipated future changes
  - Temporary unanticipated changes
- Household production is an important part of labour supply decisions
  - People allocate time between work, household production, and leisure
  - Changes in prices of goods and time costs affect these choices
- Fertility decisions are affected by income, prices, tastes, and technology
- Retirement decisions are affected by age, wealth, health, family, and pensions